

A Short Note on Endianness

Byte order decides how a multibyte integer is laid out in memory.

Big-endian stores the most significant byte at the lowest address.

Little-endian stores the least significant byte first, as x86 does.

Network protocols standardise on big-endian, called network order.

So htonl and ntohl convert between host order and that convention.

Why it bites you

Serialising a struct with raw memcpy bakes in the writer's byte order.

A reader on the other endianness then decodes every field backwards.

The fix is an explicit wire format, not a mirror of process memory.